

Topics in Error-Correcting Codes

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Lecture 5: Interactive Coding (September 23, 2026)

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Interactive Coding

Question: Can two parties interact over a noisy channel?

Example: Two players playing online chess over noisy channel.

We start by defining the notion of a binary protocol. In what follows we assume for simplicity that in each round both

Definition 1 (Binary protocol). In a binary protocol, Alice takes as input a sequence of functions $\Pi_A = \{\Pi_A^i\}_{i \in [\text{odd}]}$, where

$$\Pi_A^i : \{0, 1\}^{i-1} \rightarrow \{0, 1\} \cup \{\perp\},$$

and where $\perp$ means the end of the interaction. Similarly, Bob takes as input a sequence of functions $\Pi_B = \{\Pi_B^i\}_{i \in [\text{even}]}$, where

$$\Pi_B^i : \{0, 1\}^{i-1} \rightarrow \{0, 1\} \cup \{\perp\}.$$

The output of the binary protocol (Π_A, Π_B) is a sequence $(w_1, \dots, w_k)$ such that:

- For every odd $i \in [k]$, $\Pi_A^i(w_1, \dots, w_{i-1}) = w_i$.
- For every even $i \in [k]$, $\Pi_B^i(w_1, \dots, w_{i-1}) = w_i$.
- If k is even then $\Pi_A^i(w_1, \dots, w_k) = \perp$, and if k is odd then $\Pi_B^i(w_1, \dots, w_k) = \perp$.

Remark. Note that in the above we assume that each party sends a single bit per round. This is without loss of generality since we can send one bit at a time, at the cost of increasing the communication complexity by a factor of 2. This of course, does not preserve the round complexity, but for now we will not worry about preserving the round complexity.

Remark. Note that the above assumes that the protocol is deterministic.¹

From now on we assume that the number of rounds, denoted by k is fixed (padding with dummy bits if necessary). Thus, we can associate with each such protocol a binary depth k protocol tree.

Schulman [1] was the first to study the question of error correction in the interactive setting.

¹ This is without loss of generality since we can fix the parties' randomness.

Definition 2 (α -Error-Resilient Interactive Coding). An α -error-resilient interacting coding scheme converts any protocol (Π_A, Π_B) into a new protocol (σ_A, σ_B) together with two functions (f_A, f_B) such that the following holds: Fix any vectors $(a_1, \dots, a_k)$ and $(b_1, \dots, b_k)$ for which:

- For every odd $i \in [k]$, $a_i = \sigma_A(a_1, \dots, a_{i-1})$.
- For every even $i \in [k]$, $b_i = \sigma_B(b_1, \dots, b_{i-1})$.
- $|\{i \in [k] : a_i \neq b_i\}| \leq \alpha k$.

Then $f_A(a_1, \dots, a_k) = f_B(b_1, \dots, b_k) = \text{Output}(\Pi_A, \Pi_B)$.

We note that while we assume that the underlying protocol (Π_A, Π_B) each message is a single bit, this is not the case in the error-resilient protocol (σ_A, σ_B) . In the latter each a_i and b_i is a string of bits of constant size.

Definition 3 (Interactive Coding). An interactive coding scheme is one that is ϵ -error resilient for a constant $\alpha \in (0, 1)$ and where the number of bits communicated in (σ_A, σ_B) is only a constant times larger than the number of bits communicated in (Π_A, Π_B) .

Interactive Coding Constructions

Attempt 1: Encode each message in the protocol using a (non-interactive) error correcting code.

Drawback: This only corrects errors locally. It is good for random errors by repeating each sent bit $\omega(\log k)$ times.

Our focus is on adversarial error. So we need a **global** error-correction mechanism.

Tree Codes

We now define the notion of a tree code, first defined by Schulman [2]. This is an online code which can be used for encoding binary protocols.

Definition 4. A tree code, T , is an online code with,

$$T : \{0, 1\}^* \rightarrow \Sigma^*,$$

such that the following conditions hold:

- $|x| = |y| \implies |T(x)| = |T(y)|$.
- For every $x \in \{0, 1\}^*$, $T(x)$ is the prefix of $T(x\|0)$ and $T(x\|1)$.
- There exists a constant $c \in \mathbb{N}$ such that $|\Sigma| = 2^c$.²

Note that by the definition of a tree code, for every $k \in \mathbb{N}$, every $x \in \{0, 1\}^k$, and every $i \in [k]$, $T(x_1, \dots, x_k)_{[1:i]}$ depends only on $(x_1, \dots, x_i)$.³

Definition 5. The rate of a tree code $T : \{0, 1\}^* \rightarrow \Sigma^*$, where $\Sigma = 2^c$, is $R = \frac{1}{c}$.

² Without loss of generality we think of

$\Sigma = \{0, 1\}^c$.

³ Here and below, a subscript $[j : t]$ on an encoded or received string denotes the concatenation of its output blocks $j, \dots, t$.

How should we define distance? Take any two conversations $x, y \in \{0, 1\}^k$. Suppose that $(x_1, \dots, x_{i-1}) = (y_1, \dots, y_{i-1})$ then

$$T(x)_{[1:(i-1) \cdot c]} = T(y)_{[1:(i-1) \cdot c]}$$

Thus the relative distance of a tree code is defined as follows:

Definition 6. A rate R tree code T has relative distance δ if

$$\forall x, y \in \{0, 1\}^k, \forall i \in [k], \quad x_i \neq y_i \implies \Delta(T(x), T(y)) \geq \frac{\delta(k-i+1)}{R}.$$

Theorem 7 (Tree codes imply interactive coding). *Suppose that T is a tree-code of rate $R = 1/c$ and relative distance $\delta > 0$, where c is a positive integer. Then there is a generic deterministic transformation taking any binary protocol communicating k bits into a protocol communicating $12k/R$ bits, such that both parties recover the original transcript even if an adversary corrupts a $\delta/96$ fraction of the transmitted bits.*

At a high level, each party maintains a tentative vertex in the protocol tree of the underlying binary protocol, and uses a tree code to communicate its moves, together with the bit prescribed by its input at its current vertex. By decoding the received messages, each party obtains a guess for the other party's current vertex. If it believes that their vertices coincide, it moves down the tree according to the bit supplied by the owner of that vertex. If it believes that it is ahead of the other party on the same path, or that their vertices lie on different branches, it backtracks. Otherwise, it believes that it is behind the other party on the same path, and stays. Each move, including backtracking or staying, is reported in the next round. Importantly, the tree-code history only grows: backtracking in the protocol tree is recorded as a new move, rather than erasing a previous move.

When both parties decode correctly, they either advance together along the correct path or reduce the disagreement between their tentative paths. Incorrect decoding can disrupt this progress, but the tree-code distance guarantee allows us to charge such disruptions to channel corruptions. Consequently, as long as the fraction of corrupted bits is sufficiently small, the parties make enough progress to recover the original transcript.

Proof. Set

$$N = 2k, \quad L = 3c = \frac{3}{R}, \quad \alpha = \frac{\delta}{3}.$$

N is the number of rounds in our interactive coding scheme, L is the length of each message sent. Our scheme uses the following tree code, over $\Sigma = \{0, 1\}^L$, with relative distance α .

The tree code. Let

$$\mathcal{A} = \{0, 1, \text{stay}, \text{back}\}, \quad \Sigma = \mathcal{A} \times \{0, 1\},$$

and fix a bijection $e : \Sigma \rightarrow \{0, 1\}^3$. Define the online code

$$\tilde{T}(h_1, \dots, h_t) = T(e(h_1) \parallel \dots \parallel e(h_t)).$$

This code emits L bits per input symbol, we thus view its output as a sequence of L -bit blocks.

We first verify that $\tilde{T}$ has relative bit distance at least α . Fix $h, h' \in \Sigma^t$ whose first disagreement is at position i . The binary strings

$$e(h_1) \parallel \cdots \parallel e(h_t) \quad \text{and} \quad e(h'_1) \parallel \cdots \parallel e(h'_t)$$

first disagree at some position $j \leq 3i$. Moreover, by the online property, $\tilde{T}(h)$ and $\tilde{T}(h')$ agree on their first $i - 1$ output blocks. Therefore,

$$\begin{aligned} \Delta(\tilde{T}(h)_{[i:t]}, \tilde{T}(h')_{[i:t]}) &= \Delta(\tilde{T}(h), \tilde{T}(h')) \\ &\geq \delta c(3t - j + 1) \\ &\geq \delta c(3(t - i) + 1) \\ &\geq \alpha L(t - i + 1). \end{aligned} \tag{1}$$

The protocol tree. Let Π be the original protocol, and let $\mathcal{P}$ be its binary protocol tree. Each internal vertex is owned by one party, and the owner's input determines the outgoing bit.

We assume that every transcript has length k (padding with dummy bits if necessary). We further extend each leaf by a full binary tree so that the resulting tree has depth N . At every added internal vertex, Alice is the owner and her prescribed bit is 0. Recovering the first k bits of the extended execution suffices to recover the original transcript.

Identify every vertex with its root-to-vertex bit string. Write ε for the root, $|v|$ for the depth of v , and $\text{par}(v)$ for its parent, with

$$\text{par}(\varepsilon) = \varepsilon.$$

For $P \in \{A, B\}$, let

$$b_P(v) = \begin{cases} \text{the bit prescribed by } P\text{'s input at } v, & \text{if } P \text{ owns } v, \\ 0, & \text{otherwise.} \end{cases}$$

Define the move operation by

$$\text{Move}(v, a) = \begin{cases} v0, & a = 0, \\ v1, & a = 1, \\ v, & a = \text{stay}, \\ \text{par}(v), & a = \text{back}. \end{cases}$$

For a history

$$h = ((d_1, \beta_1), \dots, (d_t, \beta_t)) \in \Sigma^t,$$

$\text{vertex}(h)$ is a vertex on the protocol tree, defined by recursion, as follows:

$$u_0 = \varepsilon, \quad u_j = \text{Move}(u_{j-1}, d_j) \quad \text{for } j \in [t],$$

and set

$$\text{vertex}(h) = u_t, \quad \text{Bit}(h) = \beta_t.$$

The simulation. At each round t , we associate with each party $P \in \{A, B\}$ a vertex v_P^t , and we denote by a_P^t the move it makes at round t . Initialize

$$v_P^0 = \varepsilon, \quad a_P^0 = \text{stay}.$$

Each party also maintains an initially empty history over Σ .

The simulation consists of N rounds, where in each round both parties transmit a message simultaneously. In round t , party P , with the other party denoted by Q , performs the following steps:

1. Append

$$h_P^t = (a_P^{t-1}, b_P(v_P^{t-1}))$$

to its history,⁴ and send the t -th output block of

$$\tilde{T}(h_P^1, \dots, h_P^t).$$

This block consists of L bits.

2. Compute (a_P^t, v_P^t) from the (possibly corrupted) history blocks that it received from the other party, as follows:

- (a) Let $z_Q^t \in \{0, 1\}^{L \cdot t}$ be the concatenation of all blocks received from Q through round t . Compute

$$\hat{h}_Q^{(t)} \in \arg \min_{h \in \Sigma^t} \Delta(\tilde{T}(h), z_Q^t),$$

breaking ties according to a fixed rule. Set

$$\hat{v}_Q = \text{vertex}(\hat{h}_Q^{(t)}), \quad \hat{b}_Q = \text{Bit}(\hat{h}_Q^{(t)}).$$

- (b) Choose a_P^t as follows:

- If $v_P^{t-1} = \hat{v}_Q$, then set $a_P^t = b_P(v_P^{t-1})$ if P owns this vertex, and set $a_P^t = \hat{b}_Q$ otherwise.
- If v_P^{t-1} is a proper ancestor of $\hat{v}_Q$, set $a_P^t = \text{stay}$.
- Otherwise, set $a_P^t = \text{back}$.

Then update

$$v_P^t = \text{Move}(v_P^{t-1}, a_P^t).$$

Note that the histories are append-only: a move back changes the protocol-tree vertex but does not erase any history symbol.

After round N , each party outputs the first k bits of its vertex, or outputs failure if its vertex has depth less than k . Any padding is then removed.

Each round communicates $2L$ bits, so the total communication is

$$M = 2LN = 6cN = 12ck. \quad (2)$$

Thus, there is a $12c$ overhead in communication complexity; in other words, the rate of the interactive coding scheme is $\frac{1}{12c} = \frac{R}{12}$.

We now prove correctness. Fix the inputs and an arbitrary pattern of channel corruptions. Let $\pi^* \in \{0, 1\}^N$ be the correct transcript of the extended protocol on these inputs.

⁴ Recall that $a_P^{t-1} \in \{0, 1, \text{stay}, \text{back}\}$ is the move that party P made after decoding the message that it received from Q in round $t-1$, which caused P to move to vertex v_P^{t-1} .

The potential function. For every $t \in \{0, \dots, N\}$, let w_t be the longest common prefix of v_A^t and v_B^t , and define

$$\Phi_t = |w_t| - \max\{|v_A^t| - |w_t|, |v_B^t| - |w_t|\} = 2|w_t| - \max\{|v_A^t|, |v_B^t|\}.$$

Since both parties start at the root, $\Phi_0 = 0$. Moreover, since w_t is a prefix of both vertices,

$$\Phi_t \leq |w_t|. \quad (3)$$

We say that a round t is *good* if both parties correctly decode the other party's entire history through round t , and say that it is *bad* otherwise. Let B be the number of bad rounds.

Claim 1. For every $t \in [N]$,

$$\Phi_t \geq \Phi_{t-1} - 3.$$

Moreover, if round t is good, then

$$\Phi_t = \Phi_{t-1} + 1.$$

Note that Claim 1 implies that

$$\Phi_N \geq (N - B) - 3B = N - 4B.$$

To finish the proof it suffices to prove Claim 1 and that $B \leq \frac{N}{8}$, since these together imply that $\Phi_N \geq N/2 = k$. Together with Equation (3), this implies that $|w_N| \geq k$. This finishes the proof since the common prefix w_t is always correct. This is the case since if a party P moves forward on a node v_P^{t-1} that it owns then it moves to the edge labeled by $b_P(v_P^{t-1})$, which is correct by definition.

Proof of Claim 1. We first consider a good round t . In a good round, each party correctly determines the other party's vertex and announced bit at the beginning of the round. There are three cases.

Case 1: $v_A^{t-1} = v_B^{t-1}$. Let v be this common vertex, and let P be its owner. Party P moves along the edge labeled $b_P(v)$. The other party correctly decodes this announced bit and moves along the same edge. Therefore, both parties move to the same child of v . It follows that

$$|w_t| = |w_{t-1}| + 1$$

and

$$\max\{|v_A^t|, |v_B^t|\} = \max\{|v_A^{t-1}|, |v_B^{t-1}|\} + 1.$$

Hence $\Phi_t = \Phi_{t-1} + 1$.

Case 2: One vertex is a proper ancestor of the other. Assume without loss of generality that v_A^{t-1} is a proper ancestor of v_B^{t-1} . Alice stays and Bob backs up. Since

$$|v_B^{t-1}| \geq |v_A^{t-1}| + 1,$$

the longest common prefix remains v_A^{t-1} , and the maximum depth decreases by exactly 1. Thus,

$$|w_t| = |w_{t-1}|$$

and

$$\max\{|v_A^t|, |v_B^t|\} = \max\{|v_A^{t-1}|, |v_B^{t-1}|\} - 1.$$

Again, $\Phi_t = \Phi_{t-1} + 1$.

Case 3: The vertices are incomparable. Both parties back up. Each vertex is a strict descendant of w_{t-1} . After backing up, each party either remains in its original branch below w_{t-1} or reaches w_{t-1} itself. Consequently, their longest common prefix remains w_{t-1} . Both depths decrease by 1, so

$$|w_t| = |w_{t-1}|$$

and

$$\max\{|v_A^t|, |v_B^t|\} = \max\{|v_A^{t-1}|, |v_B^{t-1}|\} - 1.$$

Therefore, $\Phi_t = \Phi_{t-1} + 1$ in this case as well.

Next consider an arbitrary round t . Each party changes its vertex by at most one edge. Therefore,

$$\max\{|v_A^t|, |v_B^t|\} \leq \max\{|v_A^{t-1}|, |v_B^{t-1}|\} + 1.$$

Also, moving by at most one edge implies that

$$|w_t| \geq |w_{t-1}| - 1.$$

Combining these inequalities gives

$$\begin{aligned} \Phi_t &= 2|w_t| - \max\{|v_A^t|, |v_B^t|\} \\ &\geq 2(|w_{t-1}| - 1) - (\max\{|v_A^{t-1}|, |v_B^{t-1}|\} + 1) \\ &= \Phi_{t-1} - 3. \end{aligned}$$

Finally, there are $N - B$ good rounds and B bad rounds. Since $\Phi_0 = 0$, summing the one-round bounds gives

$$\Phi_N \geq (N - B) - 3B = N - 4B.$$

□

It remains to prove that $B \leq N/8$. Let E be the number of corrupted bits. We prove the following claim.

Claim 2. It holds that

$$B \leq \frac{2E}{\alpha L}.$$

Recall that we assume that the adversary corrupts at most a $\delta/96$ fraction of the transmitted bits. Namely, we assume that

$$|E| \leq \frac{\delta}{96} M.$$

By Equation (2) and $\delta = 3\alpha$,

$$E \leq \frac{\delta}{96}M = \frac{\delta}{96} \cdot 2LN = \frac{\alpha LN}{16}.$$

By Claim 2,

$$B \leq \frac{2E}{\alpha L} \leq \frac{N}{8},$$

as desired. Thus, to conclude the proof it remains to prove Claim 2.

Proof of Claim 2. Fix a party $P \in \{A, B\}$. Let E_P be the number of corrupted bits transmitted by P , and let

$$\mathcal{B}_P = \left\{ t \in [N] : \widehat{h}_P^{(t)} \neq (h_P^1, \dots, h_P^t) \right\}$$

be the set of rounds in which the other party incorrectly decodes P 's history. For an interval $I \subseteq [N]$, let $E_P(I)$ denote the number of corrupted bits transmitted by P in rounds belonging to I .

Fix $t \in \mathcal{B}_P$, and write

$$h = (h_P^1, \dots, h_P^t), \quad \widehat{h} = \widehat{h}_P^{(t)}.$$

Let i be the first position at which h and $\widehat{h}$ disagree, and set

$$s = \widetilde{T}(h), \quad \widehat{s} = \widetilde{T}(\widehat{h}), \quad z = z_P^t.$$

By the definition of the decoder,

$$\Delta(\widehat{s}, z) \leq \Delta(s, z).$$

By the online property, s and $\widehat{s}$ agree on their first $i - 1$ output blocks. Subtracting the common contribution of these blocks gives

$$\Delta(\widehat{s}_{[i:t]}, z_{[i:t]}) \leq \Delta(s_{[i:t]}, z_{[i:t]}).$$

Here, subscripts $[i : t]$ refer to the concatenation of L -bit blocks $i, \dots, t$. It follows from (1) that

$$\begin{aligned} \alpha L(t - i + 1) &\leq \Delta(s_{[i:t]}, \widehat{s}_{[i:t]}) \\ &\leq \Delta(s_{[i:t]}, z_{[i:t]}) + \Delta(z_{[i:t]}, \widehat{s}_{[i:t]}) \\ &\leq 2\Delta(s_{[i:t]}, z_{[i:t]}) \\ &= 2E_P([i, t]). \end{aligned}$$

Thus, each $t \in \mathcal{B}_P$ has a witness interval $I_t = [i, t]$ satisfying

$$E_P(I_t) \geq \frac{\alpha L}{2} |I_t|. \quad (4)$$

We select a collection $\mathcal{I}_P$ of disjoint witness intervals as follows. Initially, all times in $\mathcal{B}_P$ are unmarked. Choose the largest unmarked time t , add I_t to $\mathcal{I}_P$, and mark every time in $\mathcal{B}_P \cap I_t$. Repeat until all times in $\mathcal{B}_P$ are marked.

After selecting an interval $I_t = [i, t]$, every remaining unmarked time is strictly smaller than i . Every subsequently selected interval therefore ends before i . Hence the intervals in $\mathcal{I}_P$ are pairwise disjoint. Moreover, they cover $\mathcal{B}_P$. Consequently, by (4),

$$|\mathcal{B}_P| \leq \sum_{I \in \mathcal{I}_P} |I| \leq \frac{2}{\alpha L} \sum_{I \in \mathcal{I}_P} E_P(I) \leq \frac{2E_P}{\alpha L}.$$

Since the set of bad rounds is $\mathcal{B}_A \cup \mathcal{B}_B$, and $E = E_A + E_B$, we obtain

$$B \leq |\mathcal{B}_A| + |\mathcal{B}_B| \leq \frac{2E_A}{\alpha L} + \frac{2E_B}{\alpha L} = \frac{2E}{\alpha L}.$$

□

□

We still need to argue that tree codes exist. We will discuss this in the next lecture.

References

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